

ANDROMEDA

Type: *Spiral*

Distance: 2,450,000 light years

Our closest large galaxy, Andromeda—a central hub surrounded by a flat, rotating disc of stars, gas, and dust—can sometimes be seen from Earth with the naked eye. In 4.5 billion years, Andromeda is expected to collide with the Milky Way, forming one huge elliptical galaxy.



CARTWHEEL GALAXY

Type: *Ring (irregular)*

Distance: 500 million light years

The Cartwheel Galaxy started out as a spiral. However, 200 million years ago it collided with a smaller galaxy, causing a powerful shock throughout the galaxy, which tossed lots of the gas and dust to the outside, creating its unusual shape.

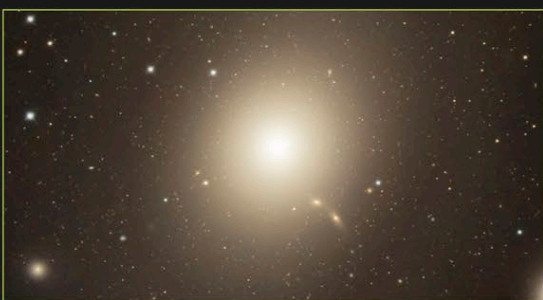


MESSIER 87

Type: *Elliptical*

Distance: 53 million light years

M87, also known as Virgo A, is one of the largest galaxies in our part of the universe. The galaxy is giving out a powerful jet of material from the supermassive black hole at its center, energetic enough to accelerate particles to nearly the speed of light.

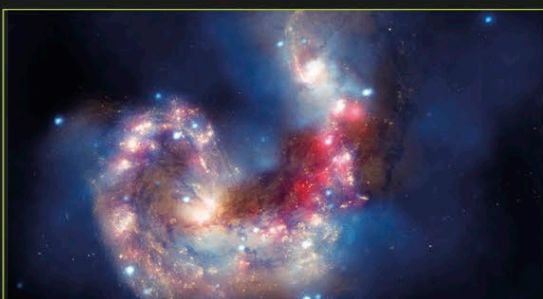


ANTENNAE GALAXIES

Type: *Merging spirals*

Distance: 45 million–65 million light years

Around 1.2 billion years ago, the Antennae Galaxies were two separate galaxies: one barred spiral and one spiral. They started to merge a few hundred million years ago, when the antennae formed and are expected to become one galaxy in about 400 million years.

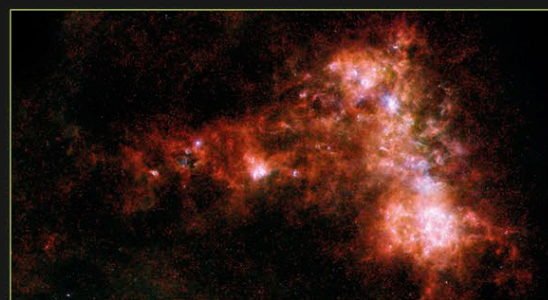


SMALL MAGELLANIC CLOUD

Type: *Dwarf (irregular)*

Distance: 197,000 light years

The dwarf galaxy SMC stretches 7,000 light years across. Like its neighbor the Large Magellanic Cloud (LMC), its shape has been distorted by the gravity of our own galaxy. Third closest to the Milky Way, it is known as a satellite galaxy because it orbits our own.



WHIRLPOOL GALAXY

Type: *Colliding spiral and dwarf*

Distance: 23 million light years

About 300 million years ago, the spiral Whirlpool Galaxy was struck by a dwarf galaxy, which now appears to dangle from one of its spiral arms. The collision stirred up gas clouds, triggering a burst of star formation, which can be seen from Earth with a small telescope.



Active galaxies

Some galaxies send out bright jets of light and particles from their centers. These “active” galaxies can be grouped into four types: radio galaxies, Seyfert galaxies, quasars, and blazars. All are thought to have supermassive black holes at their core, known as the active galactic nuclei, which churn out the jets of material.

The material near the center of the supermassive black hole is called the accretion disk. An opaque disk of dust and gas gathers around it.

